Hospital ER Performance Analysis

1. Project Overview

This project analyzes emergency room (ER) performance for a hospital using patient-level data on 9,216 ER visits between April 2023 and October 2024. The goal is to give hospital stakeholders a clear, data-driven view of patient volume, wait times, satisfaction and referrals, so they can make better decisions on staffing, resource allocation and patient care.

The project answers the question: how can the hospital use ER patient data to track performance, identify bottlenecks and improve patient care and resource planning? The deliverable is an interactive four-page Power BI dashboard (Monthly View, Consolidated View, Patient Details and Key Takeaways), supported by this report.

2. Dataset Summary

- Rows: 9,216 – one row per ER visit, with 9,216 unique Patient Ids
- Columns: 12 in the raw extract (Raw_Data.csv) and 14 in the prepared source file loaded into Power BI
- Key Features:
- Patient details (Patient Id, First Initial, Last Name, Gender, Age, Race)
- Visit details (Admission Date and Time, Admission Flag, Department Referral)
- Performance measures (Wait Time in minutes, Patient Satisfaction Score from 0 to 10)
- Period: 1 April 2023 to 30 October 2024 (579 days, 19 months)
- Missing Data: Patient Satisfaction Score is blank for 6,699 visits (72.7%); Department Referral is recorded as "None" for 5,400 visits (no referral)
- Patients CM: an undocumented 0/1 column (480 visits flagged 1) that is not used in the analysis

3. Data Cleaning and Preparation using Power Query

We prepared the ER data for analysis in Power BI as follows:

- **Data Loading:** Connected Power BI to the prepared source file (Hospital ER_Hospital ER.csv, 9,216 rows and 14 columns) using a UTF-8, comma-delimited CSV connection and promoted the first row to column headers.

- **Data Type Standardization:** Set Patient Id, first initial, last name, gender, race, department referral and the two status columns to Text; Patient Age, Patient Satisfaction Score, Patient Waittime and Patients CM to Whole Number; Patient Admin Flag to True/False; and Patient Admission Date to Date/Time. The raw dd-mm-yyyy hh:mm date text converts for all 9,216 rows with no errors.

- **Prepared Source File:** The 12-column raw extract was extended into the source file with Admitted vs Not Admitted and Target Status columns, and Patient Admission Flag is

named Patient Admin Flag. Both added columns agree with the calculated columns built in Power BI (Admission Status and Waittime Status) on all 9,216 rows.

- **Duplicates and Identifiers:** Checked for duplicate rows and duplicate Patient Ids; none were found (9,216 unique IDs, all in the 000-00-0000 format), so no rows were removed.

- **Missing Values:** The raw dataset contains blank values in two fields. Department Referral has 5,400 blank records, which were replaced with "None" in the prepared dataset to represent patients without a referral. Patient Satisfaction Score is blank for 6,699 visits (72.7%). These values were left blank rather than estimated, so satisfaction averages are calculated using only the 2,517 recorded responses.

- **Validity Checks:** Confirmed that ages run from 1 to 79, wait times from 10 to 60 minutes and satisfaction scores from 0 to 10, that every admission date falls between 1 April 2023 and 30 October 2024, and that text columns have no stray spaces. No negative, out-of-range or inconsistent values were found, so no values needed correcting.

- **Calculated Columns:** Added Patient Date, Patient Admission Time and Admission Hour from the admission timestamp; Full Gender (M, F and NC decoded to Male, Female and Non-Conforming); Admission Status (Admitted or Not Admitted); Age Group (10-year bands from 0-9 to 70-79); Waittime Status (Within Target if the wait is 30 minutes or less, otherwise Target Missed); Waittime Interval (the 2-hour arrival bands used in the day-and-hour matrix); and Patient Name (first initial and last name).

- **Calendar Table:** Created a Calendar Table from the first to the last patient date (579 days) with Year, Month, Month & Year, Month Number, Week Day and Day Name columns, used by the monthly and date-range filters and by the trend charts.

- **Measures:** No of Patients counts distinct Patient Ids; Avg Wait Time averages Patient Waittime and shows it in minutes; No of Patients Referred counts visits where Department Referral is not "None"; and the satisfaction score is the average of the recorded scores.

4. KPIs and Business Analysis

We calculated the KPIs required by the BRD and analyzed ER performance by month, wait time, day, arrival hour, patient demographics, referral department and admission status. Differences in average wait time between groups were tested with one-way ANOVA and differences in the share of patients seen within 30 minutes with chi-square tests, both at a 5%

significance level; patient counts across days, hours and months were tested against an even spread with chi-square goodness-of-fit tests.

1. **Overall KPIs** – Calculated the BRD KPIs across all 9,216 visits.

KPI	Definition	Result
No. of Patients	Distinct Patient Ids	9,216
Avg. Wait Time	Average Patient Waittime (median 35 min)	35.3 min
Patient Satisfaction Score	Average of recorded scores (2,517 responses, 27.3% of patients)	4.99
No. of Patients Referred	Visits with a department referral (41.4% of patients)	3,816
Seen Within 30 Mins	Wait of 30 minutes or less	3,749 (40.7%)
Admitted	Admission flag TRUE	4,612 (50.0%)

2. **Monthly Trend** – Compared volume, wait, satisfaction and referrals across the 19 months.

Month	Patients	Avg wait (min)	Avg satisfaction	Referred	Within 30 min
Apr 2023	479	34.9	5.30	216	43.0%
May 2023	480	34.4	5.16	189	44.4%
Jun 2023	506	35.6	5.18	201	38.5%
Jul 2023	464	34.7	4.99	184	42.5%
Aug 2023	494	36.4	5.06	191	37.9%
Sep 2023	469	34.3	4.98	209	45.2%
Oct 2023	493	34.9	4.74	211	43.0%
Nov 2023	464	35.2	5.09	212	42.5%
Dec 2023	489	34.8	4.68	211	41.5%
Jan 2024	513	36.3	4.96	214	38.4%
Feb 2024	431	36.7	4.72	179	34.3%
Mar 2024	506	35.9	5.33	195	38.3%
Apr 2024	469	35.0	4.63	185	42.9%
May 2024	519	35.8	5.15	206	37.6%
Jun 2024	485	35.5	4.71	194	38.8%
Jul 2024	488	35.2	4.79	198	41.0%
Aug 2024	530	35.1	5.18	223	39.1%
Sep 2024	466	35.1	4.91	193	40.3%
Oct 2024	471	34.1	5.31	205	44.4%

3. **Wait Time Distribution** – Grouped patients into 10-minute wait bands and compared satisfaction in each band.

Wait band	Patients	Share	Scores recorded	Avg satisfaction
10-20 min	1,947	21.1%	542	5.21
21-30 min	1,802	19.6%	467	4.84
31-40 min	1,784	19.4%	483	4.82
41-50 min	1,828	19.8%	509	5.09
51-60 min	1,855	20.1%	516	4.97

4. **Patients by Day of Week** – Compared volume and waits for each day of the week.

Day	Patients	Share	Avg wait (min)	Within 30 min
Monday	1,314	14.3%	35.7	39.0%
Tuesday	1,305	14.2%	35.1	41.2%
Wednesday	1,260	13.7%	35.2	40.8%
Thursday	1,332	14.5%	34.9	42.2%
Friday	1,310	14.2%	35.1	42.2%
Saturday	1,377	14.9%	35.6	39.2%
Sunday	1,318	14.3%	35.2	40.2%

5. **Patients by Arrival Time** – Compared volume and waits across 2-hour arrival bands.

Arrival hours	Patients	Share	Avg wait (min)	Within 30 min
00:00–01:59	778	8.4%	34.2	42.9%
02:00–03:59	769	8.3%	36.3	37.8%
04:00–05:59	769	8.3%	35.6	39.4%
06:00–07:59	790	8.6%	35.2	42.0%
08:00–09:59	779	8.5%	35.9	39.8%
10:00–11:59	747	8.1%	36.0	39.0%
12:00–13:59	776	8.4%	34.8	41.8%
14:00–15:59	769	8.3%	34.6	43.0%
16:00–17:59	730	7.9%	34.6	41.8%
18:00–19:59	753	8.2%	34.5	43.2%
20:00–21:59	760	8.2%	35.0	41.7%
22:00–23:59	796	8.6%	36.4	35.9%

6. **Patients by Age Group** – Compared volume and waits across 10-year age bands.

Age group	Patients	Share	Avg wait (min)	Within 30 min
0-9	1,056	11.5%	35.0	40.3%
10-19	1,179	12.8%	35.5	40.2%
20-29	1,188	12.9%	35.4	40.3%
30-39	1,200	13.0%	35.8	39.3%
40-49	1,135	12.3%	35.1	41.1%
50-59	1,151	12.5%	35.0	41.4%
60-69	1,154	12.5%	34.8	42.4%
70-79	1,153	12.5%	35.4	40.4%

7. **Patients by Gender** – Compared volume and waits by gender.

Gender	Patients	Share	Avg wait (min)	Within 30 min
Male	4,705	51.1%	35.4	40.4%
Female	4,487	48.7%	35.1	41.0%
Non-Conforming	24	0.3%	37.2	20.8%

8. **Patients by Race** – Compared volume and waits by patient race.

Race	Patients	Share	Avg wait (min)	Within 30 min
White	2,571	27.9%	35.1	40.9%
African American	1,951	21.2%	35.6	39.5%
Two or More Races	1,557	16.9%	35.4	41.7%
Asian	1,060	11.5%	35.3	41.4%
Declined to Identify	1,030	11.2%	34.9	40.0%
Pacific Islander	549	6.0%	34.6	43.0%
Native American/Alaska Native	498	5.4%	35.7	38.0%

9. **Department Referrals** – Counted referrals to each department and their share of all patients and of all referrals.

Department referral	Patients	% of all patients	% of referrals
None (not referred)	5,400	58.6%	–
General Practice	1,840	20.0%	48.2%
Orthopedics	995	10.8%	26.1%
Physiotherapy	276	3.0%	7.2%
Cardiology	248	2.7%	6.5%
Neurology	193	2.1%	5.1%
Gastroenterology	178	1.9%	4.7%
Renal	86	0.9%	2.3%

10. **Admission Status** – Compared admitted and not admitted patients on volume, wait and satisfaction.

Admission status	Patients	Share	Avg wait (min)	Within 30 min	Avg satisfaction
Admitted	4,612	50.0%	35.0	41.5%	5.08
Not Admitted	4,604	50.0%	35.6	39.9%	4.91

5. Dashboard in Power BI

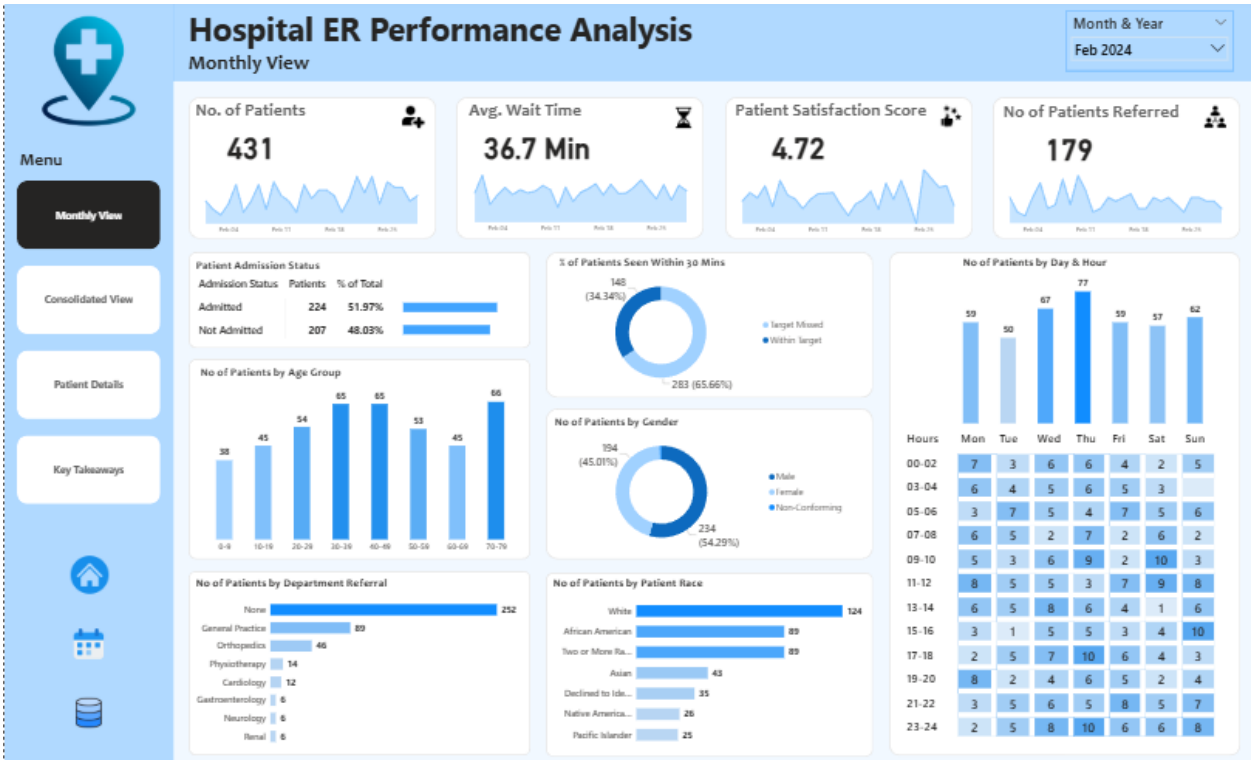
Finally, we built an interactive four-page dashboard in Power BI to present the ER performance insights visually. The pages are navigated from the menu on the left:

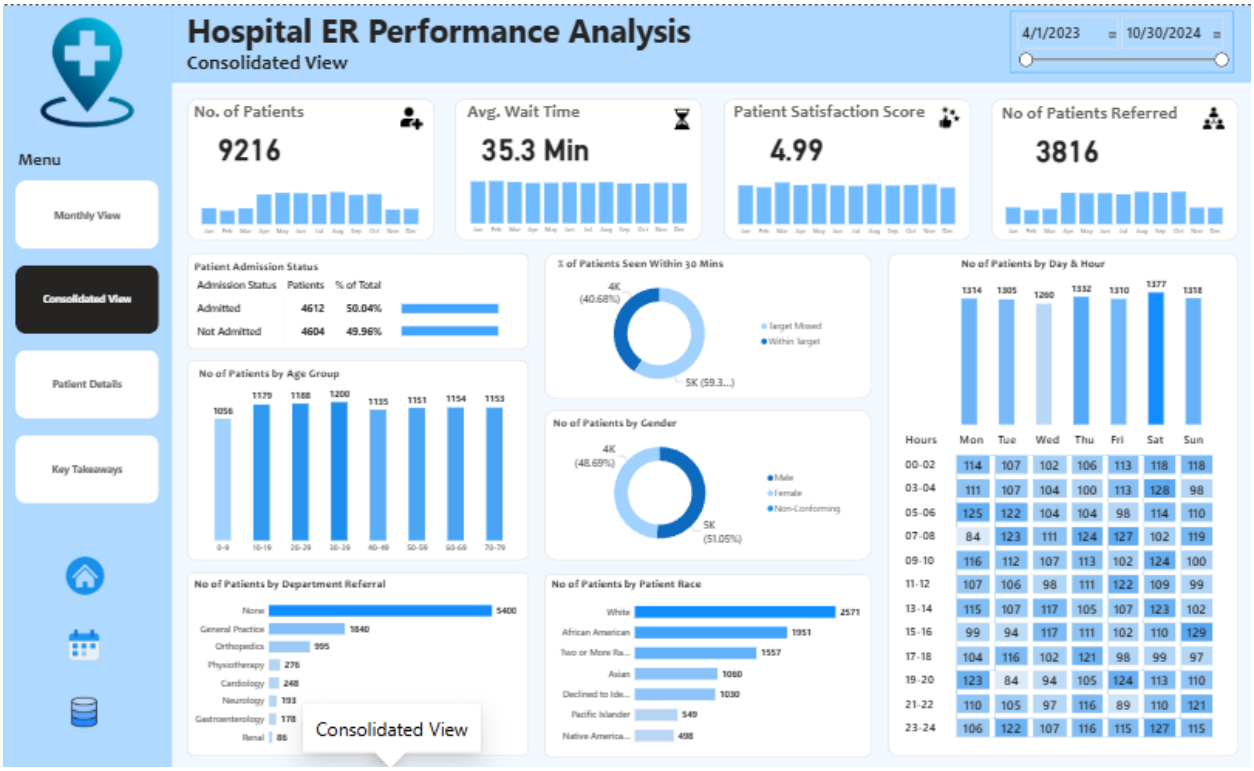
- **Monthly View:** KPI cards (No. of Patients, Avg. Wait Time, Patient Satisfaction Score and No. of Patients Referred) with daily trends for the month chosen in the Month & Year slicer, plus admission status, % seen within 30 mins, age group, gender, department referral, race and a day-and-hour matrix.

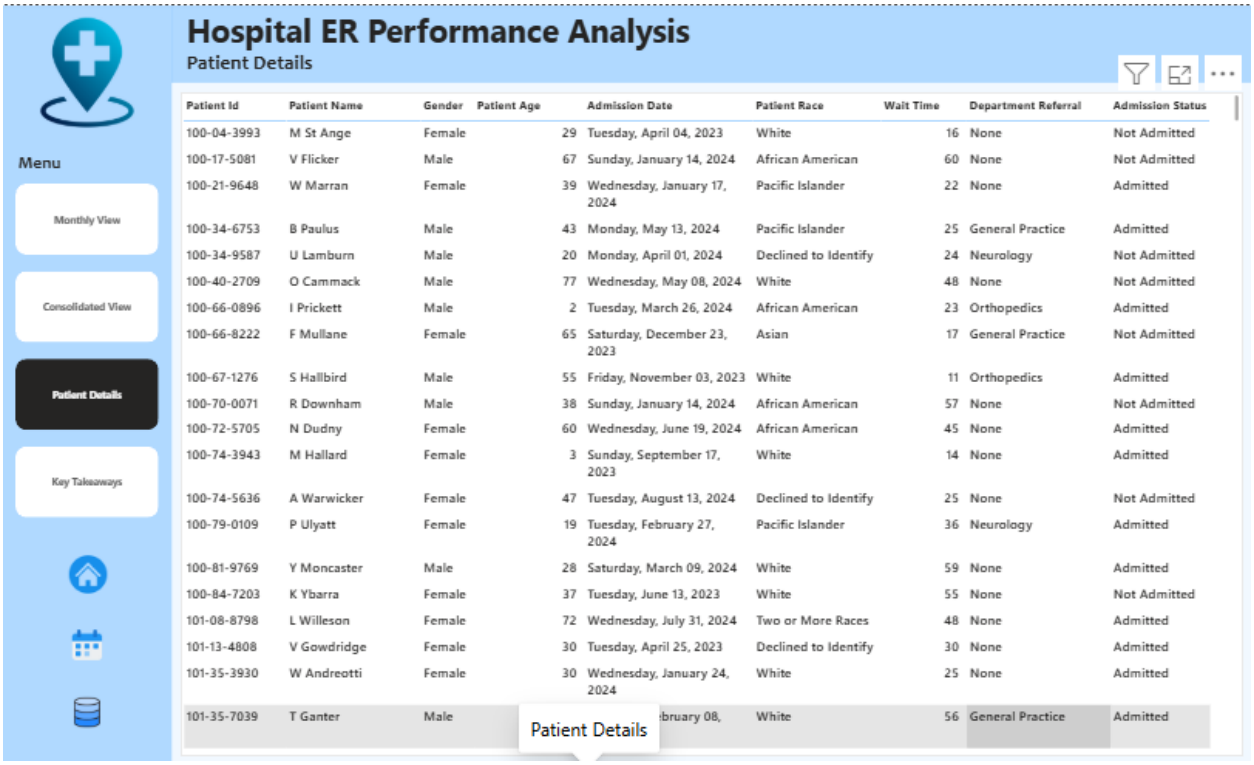
- **Consolidated View:** The same metrics over a custom date range (1 April 2023 to 30 October 2024 shown), with the trends shown by calendar month.

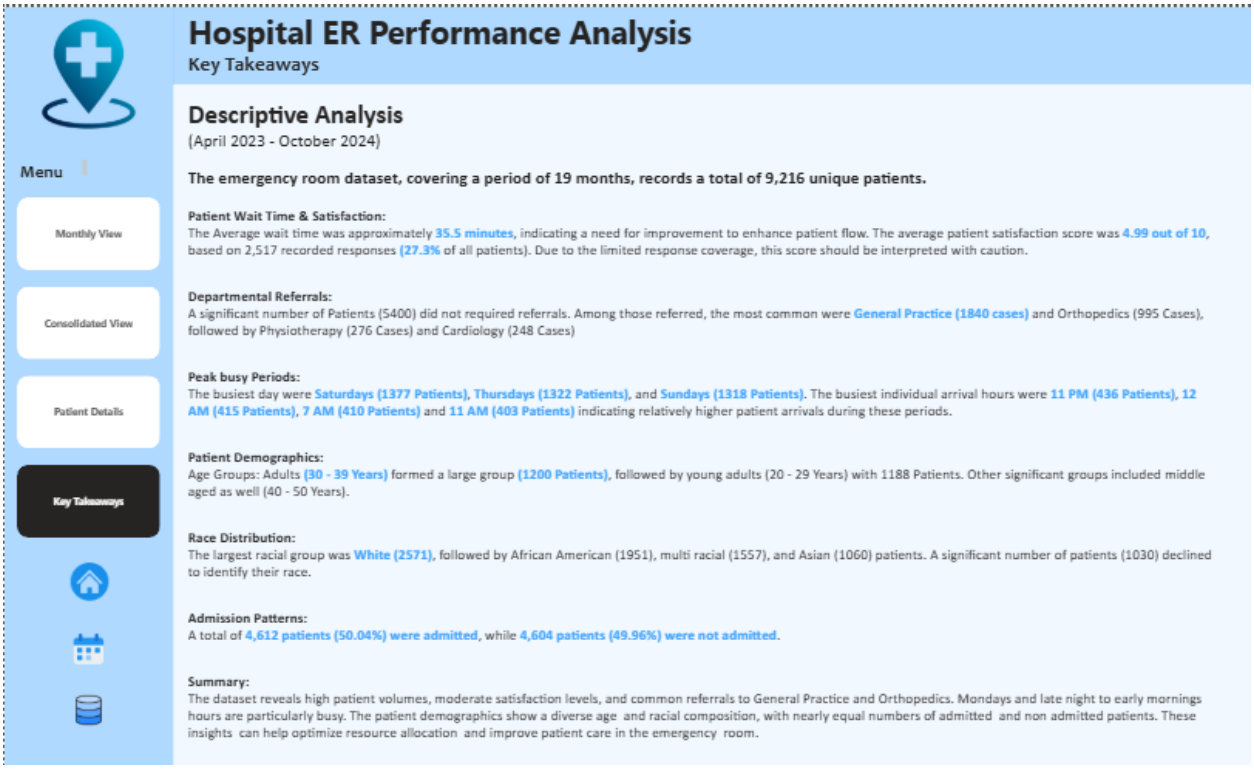
- **Patient Details:** A row-level table of every patient (ID, name, gender, age, admission date, race, wait time, department referral and admission status) for drill-down.

- **Key Takeaways:** A written descriptive summary of the main patterns.


Hospital ER Performance Analysis
Monthly View
Month & Year
Feb 2024
Menu
Monthly View
Consolidated View
Patient Details
Key Takeaways
No. of Patients
431
Avg. Wait Time
36.7 Min
Patient Satisfaction Score
4.72
No of Patients Referred
179
Patient Admission Status
Admission Status Patients % of Total
Admitted 224 51.97%
Not Admitted 207 48.03%
% of Patients Seen Within 30 Mins
148 (34.34%)
283 (65.66%)
Target Missed
Within Target
No of Patients by Day & Hour
59 50 67 77 59 57 62
Hours Mon Tue Wed Thu Fri Sat Sun
00-02 7 3 6 6 4 2 5
03-04 6 4 5 6 5 3
05-06 3 7 5 4 7 5 6
07-08 6 5 2 7 2 6 2
09-10 5 3 6 9 2 10 3
11-12 8 5 5 3 7 9 8
13-14 6 5 8 6 4 1 6
15-16 3 1 5 5 3 4 10
17-18 2 5 7 10 6 4 3
19-20 8 2 4 6 5 2 4
21-22 3 5 6 5 8 5 7
23-24 2 5 8 10 6 6 8
No of Patients by Age Group
38 45 54 65 65 53 45 66
No of Patients by Gender
194 (45.01%)
234 (54.29%)
Male
Female
Non-Conforming
No of Patients by Department Referral
None 252
General Practice 89
Orthopedics 46
Physiotherapy 14
Cardiology 12
Gastroenterology 6
Neurology 6
Renal 6
No of Patients by Patient Race
White 124
African American 89
Two or More Ra... 89
Asian 43
Declined to Ide... 35
Native America... 26
Pacific Islander 25


Hospital ER Performance Analysis
Consolidated View
4/1/2023
10/30/2024
Menu
Monthly View
Consolidated View
Patient Details
Key Takeaways
No. of Patients
9216
Avg. Wait Time
35.3 Min
Patient Satisfaction Score
4.99
No of Patients Referred
3816
Patient Admission Status
Admission Status Patients % of Total
Admitted 4612 50.04%
Not Admitted 4604 49.96%
% of Patients Seen Within 30 Mins
4K (40.68%)
5K (59.3...)
Target Missed
Within Target
No of Patients by Day & Hour
1314 1305 1260 1332 1310 1377 1318
Hours Mon Tue Wed Thu Fri Sat Sun
00-02 114 107 102 106 113 118 118
03-04 111 107 104 100 113 128 98
05-06 125 122 104 104 98 114 110
07-08 84 123 111 124 127 102 119
09-10 116 112 107 113 102 124 100
11-12 107 106 98 111 122 109 99
13-14 115 107 117 105 107 123 102
15-16 99 94 117 111 102 110 129
17-18 104 116 102 121 98 99 97
19-20 123 84 94 105 124 113 110
21-22 110 105 97 116 89 110 121
23-24 106 122 107 116 115 127 115
No of Patients by Age Group
1056 1179 1188 1200 1135 1151 1154 1153
No of Patients by Gender
4K (48.69%)
5K (51.05%)
Male
Female
Non-Conforming
No of Patients by Department Referral
None 5400
General Practice 1840
Orthopedics 995
Physiotherapy 276
Cardiology 248
Neurology 193
Gastroenterology 178
Renal 86
No of Patients by Patient Race
White 2571
African American 1951
Two or More Ra... 1557
Asian 1060
Declined to Ide... 1030
Pacific Islander 549
Native America... 498
Consolidated View


Hospital ER Performance Analysis
Patient Details
Menu
Monthly View
Consolidated View
Patient Details
Key Takeaways
Patient Id | Patient Name | Gender | Patient Age | Admission Date | Patient Race | Wait Time | Department Referral | Admission Status
100-04-3993 | M St Ange | Female | 29 | Tuesday, April 04, 2023 | White | 16 | None | Not Admitted
100-17-5081 | V Flicker | Male | 67 | Sunday, January 14, 2024 | African American | 60 | None | Not Admitted
100-21-9648 | W Marran | Female | 39 | Wednesday, January 17, 2024 | Pacific Islander | 22 | None | Admitted
100-34-6753 | B Paulus | Male | 43 | Monday, May 13, 2024 | Pacific Islander | 25 | General Practice | Admitted
100-34-9587 | U Lamburn | Male | 20 | Monday, April 01, 2024 | Declined to Identify | 24 | Neurology | Not Admitted
100-40-2709 | O Cammack | Male | 77 | Wednesday, May 08, 2024 | White | 48 | None | Not Admitted
100-66-0896 | I Prickett | Male | 2 | Tuesday, March 26, 2024 | African American | 23 | Orthopedics | Admitted
100-66-8222 | F Mullane | Female | 65 | Saturday, December 23, 2023 | Asian | 17 | General Practice | Not Admitted
100-67-1276 | S Hallbird | Male | 55 | Friday, November 03, 2023 | White | 11 | Orthopedics | Admitted
100-70-0071 | R Downham | Male | 38 | Sunday, January 14, 2024 | African American | 57 | None | Not Admitted
100-72-5705 | N Dudny | Female | 60 | Wednesday, June 19, 2024 | African American | 45 | None | Admitted
100-74-3943 | M Hallard | Female | 3 | Sunday, September 17, 2023 | White | 14 | None | Admitted
100-74-5636 | A Warwicker | Female | 47 | Tuesday, August 13, 2024 | Declined to Identify | 25 | None | Not Admitted
100-79-0109 | P Ulyatt | Female | 19 | Tuesday, February 27, 2024 | Pacific Islander | 36 | Neurology | Admitted
100-81-9769 | Y Moncaster | Male | 28 | Saturday, March 09, 2024 | White | 59 | None | Admitted
100-84-7203 | K Ybarra | Female | 37 | Tuesday, June 13, 2023 | White | 55 | None | Not Admitted
101-08-8798 | L Willeson | Female | 72 | Wednesday, July 31, 2024 | Two or More Races | 48 | None | Admitted
101-13-4808 | V Gowdridge | Female | 30 | Tuesday, April 25, 2023 | Declined to Identify | 30 | None | Admitted
101-35-3930 | W Andreotti | Female | 30 | Wednesday, January 24, 2024 | White | 25 | None | Admitted
101-35-7039 | T Ganter | Male | | bruary 08, | White | 56 | General Practice | Admitted
Patient Details


Hospital ER Performance Analysis
Key Takeaways
Menu
Monthly View
Consolidated View
Patient Details
Key Takeaways
Descriptive Analysis
(April 2023 - October 2024)
The emergency room dataset, covering a period of 19 months, records a total of 9,216 unique patients.
Patient Wait Time & Satisfaction:
The Average wait time was approximately 35.5 minutes, indicating a need for improvement to enhance patient flow. The average patient satisfaction score was 4.99 out of 10, based on 2,517 recorded responses (27.3% of all patients). Due to the limited response coverage, this score should be interpreted with caution.
Departmental Referrals:
A significant number of Patients (5400) did not required referrals. Among those referred, the most common were General Practice (1840 cases) and Orthopedics (995 Cases), followed by Physiotherapy (276 Cases) and Cardiology (248 Cases)
Peak busy Periods:
The busiest day were Saturdays (1377 Patients), Thursdays (1322 Patients), and Sundays (1318 Patients). The busiest individual arrival hours were 11 PM (436 Patients), 12 AM (415 Patients), 7 AM (410 Patients) and 11 AM (403 Patients) indicating relatively higher patient arrivals during these periods.
Patient Demographics:
Age Groups: Adults (30 - 39 Years) formed a large group (1200 Patients), followed by young adults (20 - 29 Years) with 1188 Patients. Other significant groups included middle aged as well (40 - 50 Years).
Race Distribution:
The largest racial group was White (2571), followed by African American (1951), multi racial (1557), and Asian (1060) patients. A significant number of patients (1030) declined to identify their race.
Admission Patterns:
A total of 4,612 patients (50.04%) were admitted, while 4,604 patients (49.96%) were not admitted.
Summary:
The dataset reveals high patient volumes, moderate satisfaction levels, and common referrals to General Practice and Orthopedics. Mondays and late night to early mornings hours are particularly busy. The patient demographics show a diverse age and racial composition, with nearly equal numbers of admitted and non admitted patients. These insights can help optimize resource allocation and improve patient care in the emergency room.

Independent KPI Validation – Power BI vs. Analytical Dataset

The KPI measures implemented in Power BI were independently validated against the calculated results from the prepared analytical dataset for both the Consolidated View and the Monthly View (February 2024). This reconciliation verifies the accuracy of DAX calculations, aggregation logic and dashboard outputs. Dashboard values are displayed using rounded formatting, while the analytical dataset contains the corresponding exact values.

KPI	Power BI – Consolidated	Analytical Dataset	Power BI – Feb 2024	Analytical Dataset
No of Patients	9,216	9,216	431	431
Avg Wait Time	35.3 Min	35.26 min	36.7 Min	36.67 min
Patient Satisfaction Score	4.99	4.992	4.72	4.72
Patients Referred	3,816	3,816	179	179
Admitted	4,612 (50.04%)	4,612 (50.04%)	224 (51.97%)	224 (51.97%)
Not Admitted	4,604 (49.96%)	4,604 (49.96%)	207 (48.03%)	207 (48.03%)
Seen within 30 mins	40.68%	3,749 (40.68%)	34.34% (148)	148 (34.34%)
Target missed	59.3%	5,467 (59.32%)	65.66% (283)	283 (65.66%)
Male	51.05%	4,705 (51.05%)	54.29% (234)	234 (54.29%)
Female	48.69%	4,487 (48.69%)	45.01% (194)	194 (45.01%)

6. Key Findings

- **Steady Demand** – The ER saw 9,216 patients over 19 months, about 485 a month, with no seasonal pattern: monthly volume ranges only from 431 (February 2024) to 530 (August 2024) and is not significantly different from an even spread ($p = 0.31$).

- **Wait Times Miss the 30-Minute Target** – The average wait is 35.3 minutes (median 35), and only 40.7% of patients (3,749) are seen within 30 minutes while 59.3% (5,467) miss the target. Waits are spread evenly from 10 to 60 minutes, and the monthly share seen within target stays between 34.3% (February 2024) and 45.2% (September 2023).

- **Waits Are Similar Across Patient Groups** – Average wait differs by about 2 minutes or less across days, age groups, races, admission status and referral departments, and none of these differences is statistically significant. Only arrival time shows a significant difference ($p = 0.018$): waits are longest at 22:00–23:59 (36.4 minutes) and 02:00–03:59 (36.3 minutes) and shortest at 00:00–01:59 (34.2 minutes).

- **Satisfaction Is Weakly Measured** – Only 2,517 of 9,216 patients (27.3%) have a satisfaction score. The 4.99 average is spread evenly across the 0 to 10 scale (each score makes up 8% to 10% of responses) and is unrelated to wait time (correlation −0.02), so it should be treated with caution.

- **Mild Peaks** – Saturday is the busiest day (1,377 patients, 14.9%) and Wednesday the quietest (1,260, 13.7%), with Thursday (1,332) and Sunday (1,318) next. Arrivals are spread across all 24 hours, with the highest volumes recorded at 23:00 (436 patients), 07:00 (415 patients), 13:00 (410 patients) and 00:00 406 patients). Neither the day nor the hour pattern differs significantly from an even spread ($p = 0.478$ and $p = 0.482$ respectively).

- **Referrals Concentrate in Two Departments** – 3,816 patients (41.4%) were referred. General Practice (1,840) and Orthopedics (995) account for 74.3% of referrals, followed by Physiotherapy (276), Cardiology (248), Neurology (193), Gastroenterology (178) and Renal (86).

- **Balanced Demographics and Admissions** – Patients are spread evenly across age groups from 10 to 79 (1,135 to 1,200 each, with 1,056 aged 0 to 9), and 51.1% are male, 48.7% female and 0.3% (24 patients) non-conforming. White patients are the largest racial group (27.9%) and 11.2% declined to identify. Half of the patients were admitted (4,612) and half were treated and released (4,604).

- **Data Notes** – The dataset is a historical extract (April 2023 to October 2024), and its near-uniform distributions suggest it may be simulated or sampled, so the findings should be validated with live ER data. The Patients CM column is undocumented and unused, and the 24 non-conforming patients are too few for reliable comparisons.

7. Business Recommendations

- **Reduce Wait Times Across the Board** – 59.3% of patients miss the 30-minute target on every day and in every month, which points to process-wide causes. Review triage and patient-flow steps (for example a fast-track path for low-acuity patients) and track the share seen within 30 minutes each month against a goal above 40.7%.

- **Review Overnight and Late-Evening Coverage** – Average waits are longest for arrivals between 22:00–23:59 and 02:00–03:59 (approximately 36.4 and 36.3 minutes, respectively). Individual hourly patient volumes peak at 23:00, 07:00, 13:00 and 00:00. Review staffing coverage during these periods and assess whether slightly higher coverage on Saturdays is operationally appropriate.

- **Improve Satisfaction Measurement** – Collect a satisfaction score from every discharged patient (only 27.3% have one today) and add comments, so the score can be linked reliably to waits, departments and outcomes.

- **Strengthen General Practice and Orthopedics Referral Pathways** – These two departments take 74.3% of referrals, so make sure they have the appointment capacity and fast hand-offs to absorb ER referrals, and track time to appointment for referred patients.

- **Plan Inpatient Capacity for Admissions** – Half of all ER patients (about 240 a month) are admitted, so keep admission beds and bed-management processes sized for this steady flow.

- **Investigate February 2024** – February 2024 had the lowest volume (431 patients) but the longest average wait (36.7 minutes) and the lowest share seen within 30 minutes (34.3%). Review staffing and any operational events that month.

- **Improve Data Capture and Reporting** – Record satisfaction for every visit, document the Patients CM field, add wait time by arrival hour to the dashboard, and re-check the written takeaways against the data each time the dashboard is refreshed.